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Failure of natural regeneration after clearcut logging in Wood Buffalo National Park, Canada

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Abstract

Logging of boreal riparian old-growth white spruce took place in Wood Buffalo National Park from 1951 to 1991. In this study we focus on one of the areas, Timber Berth 408, logged from 1966 to 1991. Regeneration surveys of white spruce were conducted in clearcuts and undisturbed vegetation during June 1994. The age of a clearcut is unrelated to either white spruce stocking or density over a 25 year period. Post-logging natural regeneration in clearcuts has failed: median stocking (% frequency in 10 m^2 plots) and density (stems ha^{-1}) of seedlings are 18.2% and 300 ha^{-1}, and of transgressives are 14.3% and 200 ha^{-1}. Median stocking and density of spruce (including residual growth) in the cutover areas are 36.4% and 848 ha^{-1}. In order to achieve the level of stocking found prior to logging, ≈ 93% of the logged area, or 9300 ha, would require planting. There is a clear relationship between spruce stocking rates and densities in clearcuts and the distance to the nearest white spruce seed tree. White spruce seed tree distance is important in determining the ability of a clearcut to regenerate naturally. Failure of post-logging natural regeneration in Wood Buffalo National Park is attributable primarily to two factors: (1) large size of the clearcuts, placing most of any area logged beyond the effective dispersal distance of white spruce seed; (2) destruction of advance growth and lack of residual growth. Clearcutting and site preparation are shown to degrade boreal riparian ecosystem structure and function. Recommendations are provided whose goals are to maintain biodiversity and healthy ecological structure and function in boreal riparian ecosystems.

Keywords: Forestry; Management; Old-growth; Riparian; Silviculture; Stocking; White spruce

1. Introduction

Logging in Wood Buffalo National Park, a UN World Heritage Site, was conducted from 1951 until 1991 (Olsen, 1992). There are six timber berths in Wood Buffalo National Park, all in lowland alluvial white spruce (*Picea glauca* (Moench) Voss) ecosystems. Most of the old-growth riparian white spruce forests have been logged from these berths, with the largest remnant in Timber Berth 408 (TB408) (Timoney, 1995). Within TB408, the largest of the former timber berths, clearcuts vary in size, year of cut, and amount of residual growth. From the outset of the logging, white spruce regeneration failure was noted (Wagg, 1964; Flanagan, 1992), yet nothing was done to improve the logging practices until the late 1980s (Timoney, 1995).

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Logging of TB408 began in 1966 and ended in 1991. No restoration or artificial regeneration measures have been undertaken on the logged lands. TB408 thus serves as a useful area for studying natural regeneration after clearcut logging in a boreal alluvial environment. At least 10000 ha have been logged from TB408, about two-thirds of which were cut between 1980 and 1990 (Anonymous, 1990; Olsen, 1992; Timoney, 1995). The large size of the clearcuts, loss of snags, lack of watercourse buffers, excessive debris in streams, and loss of vertical and horizontal structure are likely to have long-term detrimental impacts on wildlife (Stelfox, 1991; Timoney and Robinson, 1996).

In Canada, 87% of all logging is by clearcut methods, and 80% of all logged areas undergo mechanical site preparation (Anonymous, 1994). The effects of clearcutting are manifold. Impacts are felt at all levels of ecosystem integration from the landscape system down through landscape, stand, population, and genetic levels (Franklin and Forman, 1987; Soule, 1987). In general, traditional forestry tends to homogenize age-class structure (temporal diversity), through clearcutting, elimination of old-growth forests, and use of short rotations (Stelfox, 1995). At the stand level, simplification may reduce species diversity (especially of old-growth dependent species) and horizontal and vertical stand structure (Coates and Steventon, 1995). Managed landscapes exhibit different spatial patterns from unmanaged landscapes; for example, managed landscapes may have significantly more small forest patches that are simpler in shape than forest patches in undisturbed landscapes (Mladenoff et al., 1993). Fragmentation and loss of forest interior may precipitate a decline in 'interior' and 'area-sensitive' species, and increases in nesting failure, due to both nest predation and parasitism (Farr, 1993; Hannon, 1993; Steventon et al., 1995; Faaborg et al., 1993; Timoney and Robinson, 1996).

Stand structural and functional attributes such as: (1) large logs (which, for example, function as nursery logs, sites of nitrogen fixation, and habitat for many mosses, lichens, vascular plants, fungi, and invertebrates); (2) snags (for example, used as perches, foraging, and nesting sites for a range of birds, mammals, and invertebrates); (3) canopy height and stratal diversity and canopy gaps; (4) symbiotic mycorrhizal soil fungi; (5) large trees are typically lost or degraded through logging, particularly during site preparation (Cline et al., 1980; Maser, 1990; Hammond, 1991; Morrison and Raphael, 1993; Graham et al., 1994; Coates and Burton, 1995; Coates and Steventon, 1995; Lee et al., 1995; Lumley et al., 1995; Timoney and Robinson, 1996).

Riparian ecosystems may be the most important habitat on the landscape (Coates and Steventon, 1995). As places of high diversity, travel corridors, and as the interface between aquatic and terrestrial communities, these flood-driven ecosystems are unique in the predominantly fire-driven expanse of the boreal forest (Hunter, 1990; Coates and Steventon, 1995; Timoney and Robinson, 1996). Their unique status and restricted distribution has not provided them with special protection as riparian ecosystems are the most intensively logged type in the boreal forest of western Canada (Timoney, 1995).

This paper reports on a study of white spruce stocking and density in logged and undisturbed communities of a riparian ecosystem of the Peace River. We focus on the stand and population levels as they relate to white spruce natural regeneration after logging. Questions addressed are: (1) Is there a correlation between the age of a stand and the stocking and density of white spruce? (2) Is there a correlation between the density of white spruce trees and/or saplings and the stocking of juvenile white spruce? (3) What is the relationship between the distance to the nearest white spruce seed tree and the stocking and density of young white spruce? (4) What is the relationship between stocking and density? (5) What proportion of Timber Berth 408 must be planted with spruce in order to achieve adequate stocking and density? (6) How might boreal riparian management methods be improved to better maintain ecosystem health?

2. Study area

The study area is located in TB408, in the Peace River Lowlands of Wood Buffalo National Park, Canada (58°40′N, 113°30′W; Fig. 1). Ecosystem development and diversity are driven by periodic flooding, deposition, erosion, and channel wandering (Annas, 1977; Nanson and Beach, 1977; Peterson, 1981). Exclusive of logged communities, 71% of the forest stands studied in TB408 were of flood origin compared to 29% for fire origin forest stands; forest median stand age is 144 years ($n = 27$) (Timoney, unpublished data). Parent materials are stratified, flood-deposited alluvial silts, often with multiple buried forest floor layers in older communities; landforms are floodplains and terraces; soils are primarily moderately well-drained Orthic and Cumulic Regosols (Timoney and Robinson, 1996). Undisturbed forests exist in a continuum from pure white spruce/feathermoss, white spruce/shrub/feathermoss, and white spruce–balsam poplar, to pure balsam poplar forest (Timoney and Robinson, 1996). Logged areas undergo a succession from rose–dewberry–raspberry to balsam poplar and/or Alaska birch forest (Timoney, 1995). The study area is further described by Jeffrey (1961), Wagg (1964), and Lacate et al. (1965).

3. Methods

During the summer of 1994, 27 white spruce stocking and density surveys were conducted in

Table 1
White spruce stocking and density survey, June 1994, Timber Berth 408, Wood Buffalo National Park, Canada

Stocking no.	Seedling		T'gressive		Sapling		Tree		Seedling + T'gressive		Tree + Sapling	
	Stocking (%)	Stems ha^{-1}	Stocking (%)	Stems ha^{-1}	Stocking (%)	Stems ha^{-1}	Stocking (%)	Stems ha^{-1}	Stocking (%)	Stems ha^{-1}	Stocking (%)	Stems ha^{-1}
8	32.0	920	12.0	180	6.0	100	4.0	60	34.0	1100	6.0	140
12	2.9	86	14.3	171	0.0	0	0.0	0	17.1	257	0.0	0
13	66.0	2000	12.8	170	0.0	0	0.0	0	68.1	2170	0.0	0
4	5.0	50	35.0	650	5.0	50	0.0	0	35.0	700	5.0	50
5	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0
6	5.0	50	0.0	0	0.0	0	0.0	0	5.0	50	0.0	0
9	4.0	40	28.0	460	0.0	0	0.0	0	30.0	500	0.0	0
11	18.2	606	21.2	212	3.0	30	0.0	0	36.4	818	3.0	30
3	20.0	300	70.0	3700	0.0	0	0.0	0	85.0	4000	0.0	0
10	35.0	500	35.0	400	15.0	250	0.0	0	55.0	900	15.0	250
28	24.0	600	14.0	200	4.0	40	6.0	60	36.0	800	10.0	100
1	47.7	1318	72.7	3818	25.0	477	25.0	409	88.6	5136	36.4	727
25	4.0	40	6.0	60	0.0	0	0.0	0	10.0	100	0.0	0
21	45.2	1095	0.0	0	2.4	24	0.0	0	45.2	1095	2.4	24
2	10.0	100	0.0	0	0.0	0	0.0	0	10.0	100	0.0	0
18	90.0	3175	97.5	2475	0.0	0	0.0	0	100.0	5650	0.0	0
16	6.0	200	10.0	240	0.0	0	8.0	160	16.0	440	8.0	80
22	2.0	20	0.0	0	0.0	0	0.0	0	2.0	20	0.0	0
24	0.0	0	48.0	700	36.0	640	64.0	1180	48.0	700	70.0	1280
20	0.0	0	8.0	100	46.0	600	82.0	1640	8.0	100	88.0	1420
7	0.0	0	25.0	700	15.0	200	65.0	1250	25.0	700	75.0	850
15	16.0	280	54.0	1620	8.0	120	8.0	100	56.0	1900	14.0	200
17	32.0	720	38.0	2140	10.0	140	40.0	540	52.0	2860	46.0	540
23	22.0	260	18.0	220	8.0	80	14.0	200	36.0	480	22.0	220
19	18.0	660	34.0	1520	2.0	20	38.0	560	40.0	2180	38.0	400
26	2.0	40	10.0	140	2.0	20	12.0	120	12.0	180	12.0	140
14	30.0	1300	54.0	2260	12.0	200	40.0	560	66.0	3560	50.0	600

T'gressive, transgressive; n, number of plots; Comp., logging compartment. Veg. class: BP, balsam poplar; AB, aspen ± alaska birch ± balsam poplar; RO, rose–raspberry; SW, white spruce; MX, mixedwood; SA, willow–alder; FW, fireweed. CC: 1, clearcut; 0, undisturbed.

clearcuts ($n = 13$) and undisturbed vegetation ($n = 14$) in TB408 (Table 1). Stocking is the percentage of plots in which living spruce of a given size class are present. Density is the number of stems per hectare by size class. Circular regeneration plots of 10 m^2 (radius 1.78 m, 2.5 milacre) were used. Plot size was chosen to agree with a previous stocking study in TB408 (Anonymous, 1991), Alberta standard methods, and an a priori estimate that 1000 seedlings and transgressives ha^{-1} would constitute acceptable density (after Maass, 1991). Plots were randomly positioned (in order to avoid sampling bias) along transect lines through use of random numbers; transect bearings were changed as needed to stay within the boundaries (cutblock or forest edge, for example) of a given vegetation class. Typically 40–50 plots per site were sampled; in some cases, however, lack of time reduced the number of plots. For stand age, basal area per hectare, and total white spruce stocking rates in clearcut areas (sampled 1991), data from Anonymous (1991) were incorporated to expand the database ($n = 28$ clearcut sites). In each plot, the number of living white spruce in each size class was tallied. Size classes were: (1) seedlings (< 20 cm tall); (2) transgressives (≥ 20 cm tall, diameter at breast height (dbh) < 2.5

All		n	Date	Adj. to TBplot	Comp. no.	Basal area (m^2 ha^{-1})	Veg. class	Dist. to Sw tree (m)	CC	Stand origin	Stand age
Stocking (%)	Stems ha^{-1}										
40.0	1260	50	16.6.94	17	15	2	RO	100	1	1991	4
17.1	257	35	17.6.94	Rel08	18	0	RO	150	1	1991	4
68.1	2170	47	17.6.94	X05	18	0	RO	60	1	1991	4
35.0	750	20	15.6.94	Rel05	15	0	RO	75	1	1989	6
0.0	0	50	16.6.94	16	15	0	RO	400	1	1989	6
5.0	50	40	16.6.94	16	15	0	RO	400	1	1989	6
30.0	500	50	16.6.94	13	10	0	AB	100	1	1987	8
36.4	848	33	17.6.94	Rel07	12	0	RO	150	1	1988	7
85.0	4000	20	14.6.94	06	6	4	BP	100	1	1984	11
60.0	1150	20	16.6.94	14	10	8	AB	100	1	1969	26
40.0	900	50	29.6.94	02	1	16	BP	25	1	1970	25
95.5	6023	44	13.6.94	08	7	16	BP	25	1	1967	28
10.0	100	50	28.6.94	40	10	60	BP	150	1	1967	28
47.6	1119	42	25.6.94	34		0	FW	60	0	1990	4
10.0	100	10	14.6.94	12		0	AB	160	0	1981	13
100.0	5650	40	22.6.94	29		0	SA	300	0	1966	29
24.0	600	50	21.6.94	28		10	BP	50	0	1952	43
2.0	20	50	25.6.94	38		0	SA	100	0	1939	56
82.0	2520	50	26.6.94	39		96	MX	0	0	1930	65
90.0	2340	50	25.6.94	Rel11		44	SW	0	0	1922	73
85.0	2150	20	16.6.94	15	15	40	SW	0	0	1858	137
62.0	2120	50	21.6.94	27		14	BP	75	0	1835	160
72.0	3540	50	21.6.94	26		22	MX	0	0	1823	172
50.0	760	50	25.6.94	35		22	SW	0	0	1752	243
68.0	2760	50	23.6.94	37		28	SW	0	0	1735	260
22.0	320	50	28.6.94	Rel12	5	16	BP	50	0	1698	297
88.0	4320	50	21.6.94	33		24	SW	0	0	1657	338

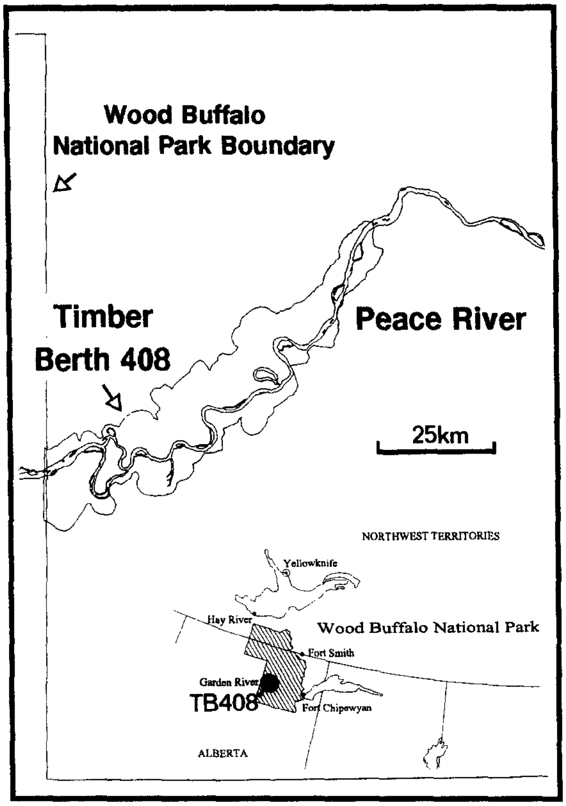

Fig. 1. The study area, which lies in the valley of the Peace River in Wood Buffalo National Park, Canada.

cm); (3) saplings (dbh 2.5–9.9 cm); (4) trees (dbh ≥ 10 cm). Transgressive, as used here, is similar to the commonly used terms 'advance regeneration' and 'advance growth' (i.e. individuals existing before logging), but differs in that much advance growth in these communities is seedling-sized. 'Total' stocking and density as used here includes all size classes of spruce. Age data from 407 spruce individuals indicated that the seedlings were 2–7 years old (range between the 10 and 90 percentiles, $n = 142$), transgressives 7–20 years old ($n = 186$), saplings 30–120 years old ($n = 24$), and trees 50–220 years old ($n = 55$). Height data from 532 spruce indicated that the seedlings were typically 2–14 cm tall (10 and 90 percentiles, $n = 142$), transgressives were 28–120 cm tall ($n = 186$), saplings were 3.2–9.3 m tall ($n = 57$), and trees were 12–32 m tall ($n = 147$).

Sites were chosen to represent a range of ages and vegetation classes both in clearcuts and natural communities. Sites were placed nearby pre-existing permanent sample plots wherever possible (Timoney and Robinson, 1996). Vegetation classes were based on dominant canopy species. Mixedwood was defined as co-dominant white spruce and balsam poplar, with neither species < 10% cover. For undisturbed communities, stand age was the basal age of the oldest tree, willow, or alder. In cutblocks, stand age was the basal age of post-logging regeneration, and was verified against cutblock maps where documentation existed. A weakness of the dataset is that no sites were sampled having stand ages of 11–24 years (logged 1971–1984; clearcuts aged 11–24 years may lie farther from the Peace River than other ages). Due to inadequate and conflicting records, it is not possible presently to determine the amount of land logged from 1971 to 1984, but it is probably about one-third of the total (Anonymous, 1990; Olsen, 1992). Two sites sampled by Tollestrup (1994), aged 11 and 15 years, were compared to help fill this age gap.

White spruce seed tree distance class was determined using ground level photographs, plot data, aerial photography, and field notes. Distance class was estimated from the survey site to the nearest individual mature white spruce tree; in most cases, the nearest individual tree was located near other spruce trees (e.g. at a forest border).

A natural regeneration survey was conducted during the summer of 1993. In that survey, growth, substrate, and competition data were gathered for 344 white spruce seedlings and transgressives. For each individual, the following were determined: height, age, root collar diameter, rooting medium (mineral, wood, duff, complex (a mixture of mineral and organic matter, often interbedded duff–silt)). In addition, vegetation data on logs and snags from permanent sample plots (unpublished data) are used to better describe the ecological differences between logged and undisturbed communities. Statistics were carried out with Systat (Wilkinson et al., 1992). Data normality were determined by a Kolmogorov–Smirnov test (Lilliefors type). In clearcuts, only seedling density, transgressive stocking, seedling + transgressive stocking, and total stocking of all size classes were distributed normally. Since most variables were non-normal, non-parametric Spearman's

Table 2
Median stocking and density values for white spruce in cutblocks compared to those of undisturbed white spruce and white spruce–balsam poplar forests, Timber Berth 408, Wood Buffalo National Park

Variable	D		G		DG		S		T		TS		All	
	%	Density	%	Density	%	Density	%	Density	%	Density	%	Density	%	Density
CC	18.2	300	14.3	200	35.0	800	0.0	0	0.0	0	0.0	0	31.4	848
SW, MX	18	260	34	700	40	700	12	200	40	560	50	600	82.0	2520
P	0.360	0.577	0.250	0.104	0.692	0.691	0.010	0.013	0.000	0.000	0.000	0.001	0.001	0.036

D, seedling; G, transgressive; S, sapling; T, tree; All, all size classes; CC, cutblock; SW, white spruce; MX, white spruce–balsam poplar mixedwood. P, Mann–Whitney probability, two-tailed; $n = 20$ sites, except for total stocking (All %, $n = 35$).

rho correlations were used to determine the significance of the relationships. Vascular plant nomenclature follows Moss (1983).

4. Results and discussion

4.1. Stocking and density in clearcuts compared to undisturbed forests

Median total spruce (all size classes) stocking and density in the timber berth cutblocks were 36.4% and 848 ha^{-1} (Table 2). For cutblocks logged from 1982 to 1991 ($n = 15$), Anonymous (1991) found a median total spruce stocking of 27.7%. Two sites with cutblock ages of 11 and 15 years sampled by Tollestrup (1994), which fill a gap in our data, had a total stocking ≈ 35–45% (cf. our 36.4% total stocking). In comparison, median total spruce stocking and density in mature and old-growth spruce and spruce–balsam poplar forests in the study area were 82.0% and 2520 ha^{-1} (Table 2). Minimum acceptable stocking in Alberta is 80% (Anonymous, 1991), in close agreement with the median natural stocking. Total stocking ($P = 0.001$) and density ($P = 0.036$) in the logged areas are significantly lower than in

Table 3
Frequency and cumulative percent for clearcuts by white spruce stocking and density percentiles, and estimated area that is ≤ to a given stocking level or median density of undisturbed riparian white spruce and mixedwood forests, Timber Berth 408, Wood Buffalo National Park

	Stocking (%)									
	10	20	30	40	50	60	70	80	90	100
Surveys (n)	4	10	13	18	22	22	26	26	27	28
Cumulative (%)	14	36	46	64	76	76	93	93	96	100
Cumulative area (ha)	1400	3600	4600	6400	7600	7600	9300	9300	9600	10000
	Stems ha^{-1}									
	250	500	750	1000	1250	1500	1750	2000	2250	2500
Surveys (n)	3	5	6	8	9	10	10	10	11	11
Cumulative (%)	23	38	46	62	69	77	77	77	85	85
Cumulative area (ha)	2300	3800	4600	6200	6900	7700	7700	7700	8500	8500

A total of 10000 ha was assumed logged; study areas $n = 28$ for stocking data, $n = 13$ for density data; 82% stocking and 2520 spruce ha^{-1} targets for 'successful regeneration' are based on median values for undisturbed riparian white spruce and mixedwood forest in the study area.

undisturbed spruce and mixedwood (Mann–Whitney U-test, $n = 35$; Table 2) indicating that post-logging natural regeneration in clearcuts has failed.

In order to achieve pre-logging median total stocking, ≈ 93% of the area, or 9300 ha, would require planting (Table 3). To achieve pre-logging median total density, 85% (8500 ha) would require planting (Table 3; median tree + sapling stocking and density in these undisturbed forests are 50% and 600 ha^{-1}). Minimum total stocking and density observed in the undisturbed spruce and mixedwood forests are 50% and 760 ha^{-1}. To achieve these

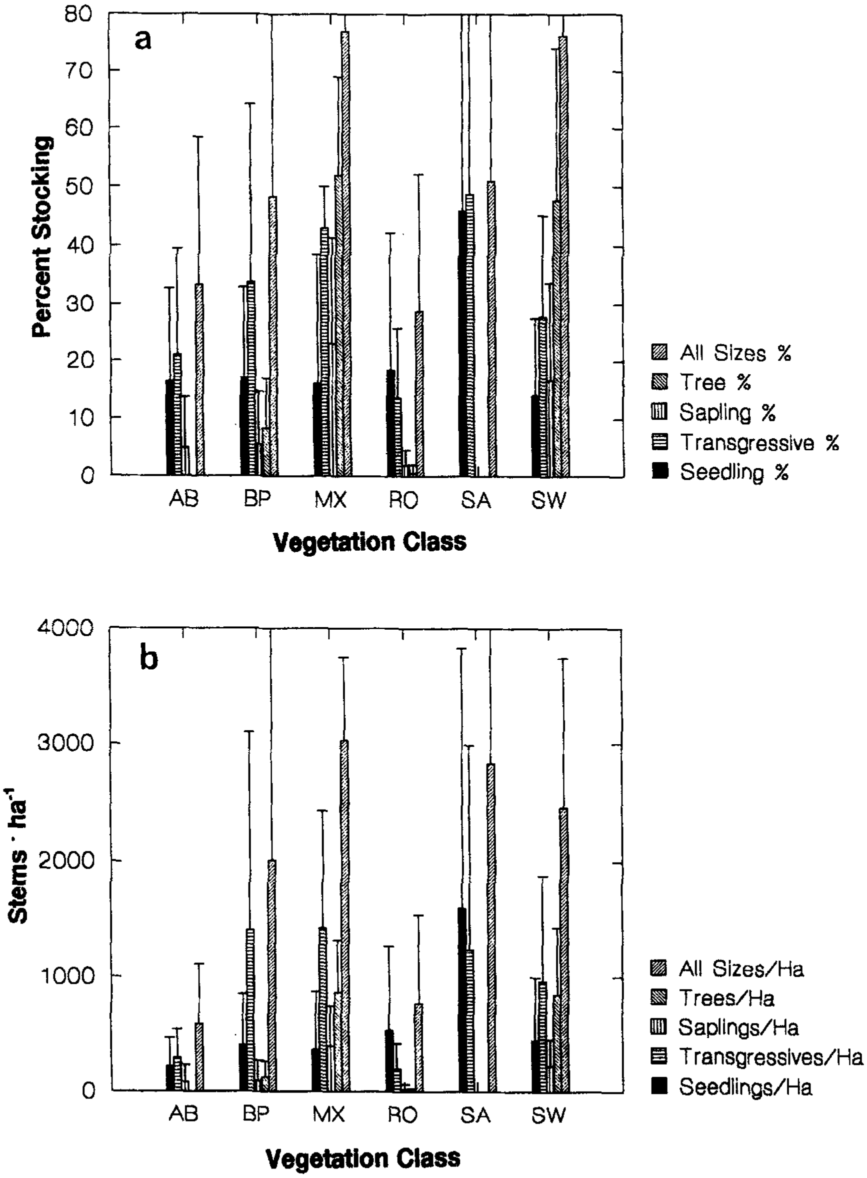

Fig. 2. Mean (±SE) white spruce stocking and density, summarized by size and vegetation classes. AB, aspen ± Alaska birch ± balsam poplar; BP, balsam poplar; MX, mixedwood; RO, rose–raspberry; SA, willow–alder; SW, white spruce.

minimum undisturbed levels, 46% (using minimum density) to 76% (using minimum stocking) of the logged areas would require planting. Heritage Canada must therefore work toward ecological restoration to fulfil its legal mandate of maintaining or restoring ecological integrity, as stated in the National Parks Act (revised 1988; Henry et al., 1995).

Natural seral communities that will succeed to spruce or mixedwood exhibit fairly high stocking and density early in succession. A 4 year old burned former old-growth white spruce forest (Table 1; Stocking no. 21), for example, had a total spruce stocking of 47.6% and density of 1119 ha^{-1}. A 28-year-old willow–alder community (Stocking no. 18) had a total stocking of 100.0% and density of 5650 ha^{-1}. In cutblocks, there are no significant correlations between cutblock age and stocking or density of any spruce size classes (Table 4). In undisturbed forests, tree and tree + sapling stocking and density are positively correlated with stand age (Table 4).

In cutblocks of TB408, over a 28-year period, the number of white spruce has not increased appreciably. These results disagree with those of Tollestrup (1994), who found a weak, though significant ($P = 0.05$), increase in spruce density over time in TB408. The discrepancy is understandable in that Tollestrup limited all his surveys to within 100 m of spruce forest edges, thereby maximizing the chance of finding young spruce (see section below on white spruce seed tree distance).

Stocking and density for each white spruce size class is summarized by vegetation class in Fig. 2(a) and Fig. 2(b). Note the low stocking and densities of spruce in the new clearcuts (RO vegetation class), and the comparable failure of regeneration in medium-aged and old clearcuts (types AB, some BP). Spruce stocking and density in undisturbed spruce, mixedwood, and willow–alder vegetation (SW, MX, SA) are provided for comparison.

4.2. Correlation between seedling and transgressive stocking and density

High correlations were observed between seedling stocking and density ($r = 0.890$), and between transgressive stocking and density ($r = 0.963$; Table 4). Future assessments of white spruce regeneration therefore need only determine the less labor-intensive stocking. Approximate densities may be derived from stocking rates by non-linear interpolation. Similar curves that predict density from stocking have been developed for parts of Oregon (Bever and Lavender, 1955) and Maritime Canada (Weetman and Frisque, 1977). A density of 1000 seedlings or transgressives ha^{-1} roughly corresponds to a natural

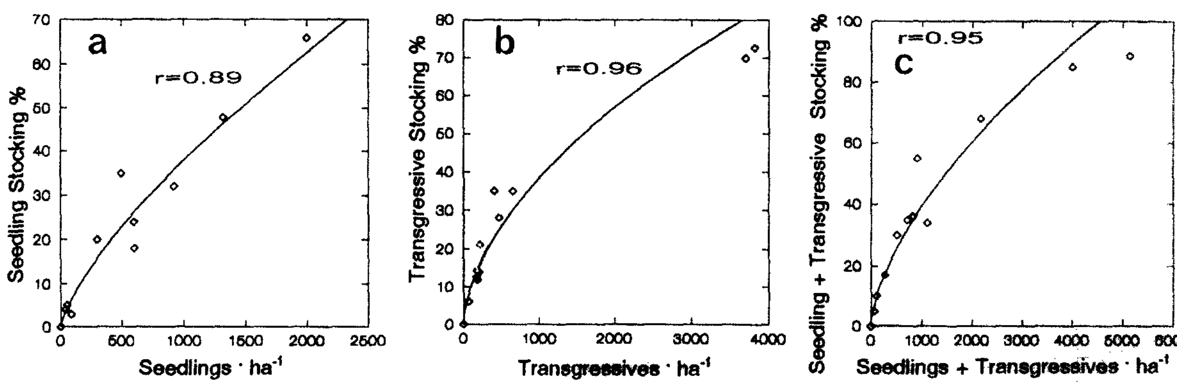

Fig. 3. Relationships between white spruce seedling and transgressive density and stocking, within size classes; clearcuts only ($n = 13$; r, rank correlation; fitted line is power curve).

Table 4
Spearman's rho correlation coefficients for stocking, density, basal area, stand age, and white spruce seed tree distance for flood- and fire-origin communities (top diagonal, $n = 14$) and in cutblocks (bottom diagonal, $n = 13$), Timber Berth 408, Wood Buffalo National Park, Canada

	D %	D density	G %	G density	D + G %	D + G density	S %	S density	T %	T density	T + S %	T + S density	All %	All density	Basal area ha^{-1}	Stand age	Sw seed tree distance
D %	***	0.968	0.310	0.396	0.654	0.693	−0.320	−0.305	−0.458	−0.475	−0.406	−0.406	0.113	0.391	−0.482	−0.033	0.249
D density	0.890	***	0.424	0.517	0.722	0.772	−0.304	−0.277	−0.405	−0.407	−0.348	−0.348	0.183	0.475	−0.420	0.064	0.224
G %	0.381	0.318	***	0.961	0.856	0.816	0.284	0.337	0.293	0.270	0.316	0.329	0.684	0.792	0.331	0.418	−0.211
G density	0.390	0.310	0.963	***	0.833	0.864	0.213	0.254	0.293	0.280	0.302	0.302	0.685	0.833	0.289	0.431	−0.239
D + G %	0.824	0.769	0.771	0.740	***	0.947	0.194	0.256	0.033	0.002	0.108	0.126	0.569	0.771	0.062	0.182	−0.066
D + G density	0.865	0.837	0.667	0.680	0.945	***	0.160	0.203	0.095	0.081	0.155	0.155	0.615	0.846	0.104	0.256	−0.153
S %	0.593	0.527	0.485	0.544	0.502	0.532	***	0.993	0.875	0.816	0.926	0.931	0.610	0.450	0.841	0.392	−0.774
S density	0.593	0.527	0.485	0.544	0.502	0.532	1.000	***	0.854	0.795	0.911	0.922	0.601	0.470	0.835	0.396	−0.757
T %	0.499	0.540	−0.164	0.294	0.342	0.457	0.646	0.646	***	0.982	0.984	0.980	0.599	0.459	0.966	0.555	−0.906
T density	0.504	0.549	0.157	0.291	0.335	0.469	0.656	0.656	0.997	***	0.968	0.963	0.590	0.449	0.971	0.516	−0.911
T + S %	0.608	0.542	0.470	0.526	0.520	0.520	0.981	0.981	0.707	0.705	***	0.996	0.620	0.483	0.959	0.536	−0.902
T + S density	0.614	0.554	0.458	0.520	0.508	0.538	0.994	0.994	0.699	0.705	0.994	***	0.616	0.492	0.968	0.528	−0.902
All %	0.880	0.811	0.698	0.694	0.968	0.982	0.538	0.538	0.469	0.470	0.553	0.553	***	0.899	0.565	0.200	−0.428
All density	0.876	0.832	0.656	0.674	0.940	0.995	0.538	0.538	0.505	0.514	0.538	0.550	0.990	***	0.452	0.279	−0.399
Basal area ha^{-1}	0.372	0.204	0.213	0.252	0.332	0.347	0.416	0.416	0.554	0.541	0.488	0.465	0.376*	0.392	***	0.563	−0.899
Stand age	0.097	−0.116	0.362	0.376	0.246	0.124	0.282	0.282	0.269	0.241	0.363	0.318	0.175*	0.141	0.285*	***	−0.630
Sw seed tree	−0.716	−0.617	−0.582	−0.636	−0.745	−0.722	−0.533	−0.533	−0.606	−0.589	−0.599	−0.568	−0.785	−0.770	−0.385	−0.161	***

D, seedling; G, *transgressive*; S, sapling; T, tree; All, all size classes; %, stocking %; density, stems ha^{-1}; basal area, basal area of living trees and saplings, m^2 ha^{-1}; Sw seed tree distance, distance to nearest white spruce seed tree; for bottom diagonal, and $P = 0.05$, $r = 0.532$; $P = 0.01$, $r = 0.661$; $P = 0.001$, $r = 0.780$; for top diagonal, and $P = 0.05$, $r = 0.514$; $P = 0.01$, $r = 0.641$; $P = 0.001$, $r = 0.760$; * for $n = 28$, at $P = 0.05$, $r = 0.370$; $P = 0.01$, $r \approx 0.470$; $P = 0.001$, $r \approx 0.580$.

stocking rate of ≈ 35% (indicative of clumped distribution). Minimum acceptable stocking in Alberta is 80% (Anonymous, 1991), and roughly corresponds to a density of ≈ 3000 seedlings ha^{-1} (Fig. 3(a)), ≈ 4000 transgressives ha^{-1} (Fig. 3(b)), and ≈ 3500 seedlings + transgressives ha^{-1} (Fig. 3(c)). Planting at 1000 seedlings ha^{-1}, however, would result in much higher stocking rates if spacing between plants is uniform or random.

4.3. Tree and sapling stocking and density compared to seedling and transgressive stocking and density

In clearcuts (Table 4), the percent stocking of seedlings is significantly correlated with sapling stocking, sapling density, tree + sapling stocking, and tree + sapling density. Spruce saplings and trees would both provide seeds and ameliorate the climate, assisting in seedling establishment. Seedling density is correlated with tree stocking, tree density, tree + sapling stocking and tree + sapling density. These relationships strengthen the notion that white spruce seedlings tend to associate with spruce saplings and trees in clearcuts.

Correlations between transgressives and tree and/or sapling stockings and densities are much weaker than those with seedlings, and largely insignificant. The lower correlations suggest that, although seedlings may establish in clearcut sites if residual adult spruce are nearby, some may not survive to the transgressive stage.

While the rank correlations observed between parent and regeneration tree stocking and density parameters are significant ($0.5 < r < 0.6$, generally), as are those relating to distance to white spruce seed tree ($r > 0.7$; see below), the correlation coefficients are not particularly high. Clearly other factors are involved, such as cutting in poor seed years, amount of exposed mineral soil, operator effort and supervision, forest conditions prior to logging, drought conditions following logging, etc. (Endean et al., 1971; Zasada, 1972; Brace, 1990).

4.4. Distance to nearest white spruce seed tree

There is a clear relationship between stocking rates and densities of clearcuts and the distance to the nearest white spruce seed tree (Table 4). Seedling + transgressive stocking (Fig. 4(a)) and density (Fig. 4(b)) show the strongest correlations with distance. All other variables also have strong relationships with distance to the nearest seed tree. Thus, the distance to the nearest white spruce seed tree is important in determining the ability of a clearcut to

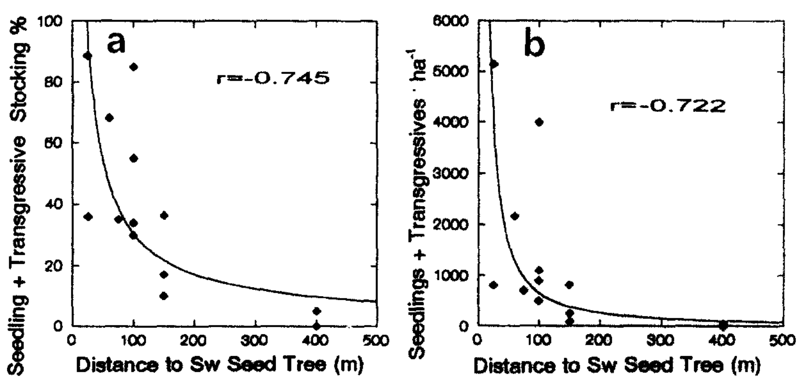

Fig. 4. Relationships between distance to nearest white spruce seed tree and stocking and density of seedlings and transgressives in clearcuts ($n = 13$; r, rank correlation; fitted line is power curve).

regenerate naturally. These results corroborate previous studies on dispersal of white spruce seed which found virtually no white spruce establishment beyond 200 m, and minimal establishment beyond 100 m (Rowe, 1955; Dobbs, 1976; Zasada, 1972). Establishment beyond 100–200 m from seed trees, observed occasionally in the study area (e.g. Stocking no. 18, Table 1), may be attributable to dispersal of seed by floodwaters or to late dispersal over snow. It is also possible that dispersal of up to 300 m may occur due to turbulence and convection (Rowe, 1955).

In TB408, a seed tree distance of 100 m corresponds roughly to a seedling + transgressive density of ≈ 700 ha^{-1} and a stocking of $\approx 30\%$ in clearcuts. At 50 m, density of spruce rises to ≈ 1500 ha^{-1} and stocking to $\approx 50\%$. Jarvis (1963) found no significant difference in white spruce seedfall in the centre-third of 40 m wide clearcut strips as compared to clearcut edges. At 50 and 200 m from forest edge, white spruce seedfall may be as low as 7% and 0.1%, respectively, of seedfall within the forest (Nienstaedt and Zasada, 1990).

4.5. Rooting medium and seedbed preferences

Median ages for spruce were 7 years (on mineral soil), 6 years (wood), 7.5 years (duff), and 10 years (complex). Spruce age class distributions were similar for mineral, wood, and duff (Kolmogorov–Smirnov (KS) test two-sample, $P = 0.086$ to 0.527, all positively skewed), whereas age distribution on the complex medium was different (KS test two-sample, $P = 0.000$ to 0.008), and marginally normal (KS test, Lilliefors one-sample, $P = 0.051$). Spruce on the complex medium were significantly older than those on other media (Mann–Whitney U-tests, $P = 0.000$ to 0.022). It is possible that spruce on the complex medium have higher survivorship; it is more likely, however, that as a seedling ages the probability of being classified as rooted in a complex medium increases due to both organic and silt accumulation.

Overall 23.5% of white spruce were rooted in mineral soil, compared to 35.8% in wood, 14.0% in duff, and 26.7% in complex ($n = 344$; Timoney and Robinson, 1996). Relative percentages for mineral soil seedlings (23.8% in undisturbed forests vs. 23.2% in non-scarified clearcuts) and duff-rooted seedlings (15.0% vs. 12.3%) are similar between site types. In contrast, wood-rooted seedlings are relatively more abundant in forests (42.2% vs. 26.1%), and complex-rooted seedlings are relatively more abundant in clearcuts (18.9% vs. 38.4%). Currently, log attributes (median density, length, diameter) do not differ between clearcuts ($n = 21$ sites) and undisturbed forests ($n = 17$ sites) (median 366 logs ha^{-1} vs. 300 ha^{-1}; 11.0 vs. 11.0 m length, 27.0 vs. 26.0 cm diameter). There was probably a pulse of log recruitment at time of logging.

Logs in cutblocks tend to be drier and lack the typical cover of mosses—this may help to explain the apparent lower suitability of wood as a rooting medium in clearcuts. Conversely, disturbance of logs by heavy machinery (even in the absence of scarification) may decrease the suitability of wood as a rooting medium, and increase the relative amount of complex media through mixing of duff, wood, and mineral soil. It is noteworthy that older cutblocks (age > 24 years, $n = 5$) have a higher median log density of 433 ha^{-1} than do younger cutblocks (< 11 years old, $n = 15$) at 267 ha^{-1}. It appears that recent logging may have destroyed more logs, or removed more trees, than did the earlier logging.

Comparison of rooting medium and medium availability for white spruce ($n = 668$) in non-scarified clearcuts (raw data from Tollestrup, 1994) allows an estimate of rooting medium 'preference': 22.1% of white spruce was rooted in mineral soil (comprising 3.2% of available surface), 28.0% in wood (12.4% of surface), 28.6% in duff (77.6% of surface), and 22.2% in complex (6.8% of surface). If preference is (% of seedlings)/(% available surface), then white spruce rooting medium preference in non-scarified clearcuts on silty alluvium of the Peace River is 6.9 mineral, 2.3 wood, 0.4 duff, 3.3 complex. It is clear that white spruce 'prefers' (i.e. is relatively more abundant) mineral soil, complex substrate, and rotted wood, and is less abundant on duff. In undisturbed forests, preferred natural seedbeds for white spruce are decayed logs and stumps, the exposed mineral soil of tip-up mounds (Jarvis et al., 1966), and on flood-deposited alluvium (Nienstaedt and Zasada, 1990). Duff is generally a poor seedbed due to desiccation, but on moist sites it may support good white spruce regeneration (Kobalinski, 1964).

While numerous studies have shown that white spruce regenerates well on scarified soil (Jarvis et al., 1966; Nienstaedt and Zasada, 1990), it does not *require* a mineral soil seed bed for establishment.

4.6. Failure of natural regeneration in TB408

Failure of post-logging natural regeneration in Wood Buffalo National Park is attributable primarily to two factors: (1) large size of the clearcuts, placing most of any area logged beyond the effective dispersal distance of white spruce seed; (2) destruction of advance growth (seedlings and transgressives) and lack of residual growth (trees and saplings).

Currently it is not possible to describe quantitatively the size distribution of clearcuts in the timber berth (an ecological land classification is needed). The clearcuts were 'progressive' (grew progressively larger with each year), and most exceed 1 km^2. The largest recent clearcut is in Compartment 15, where one continuous block exceeds 15 km^2 (1500 ha). Similar large blocks are located in Compartments 1, 8, 9, and 10.

Regeneration of white spruce on non-scarified clearcuts in TB408 was compared to that on scarified and seeded cutblocks on the Peace River bordering Wood Buffalo National Park by Tollestrup (1994). His results are difficult to interpret due to complex site histories. For example, in the scarified sites:

(1) The 4-year-old cutblocks were planted with container stock, and these seedlings were ignored; natural regeneration occurred in all 4 years, with peaks in years 2 and 3.

(2) The 10-year-old cutblocks were aerial-seeded, and these were not ignored; the 8-year-old cohort (2 years after site preparation), is the most abundant, and is from natural regeneration; natural seedlings far outnumber those aerial-seeded.

(3) The 25-year-old cutblocks were scarified and hand seeded; there was complete artificial regeneration failure, thus the site was re-scarified 3 years after logging, and re-seeded; (subsequent floods and mineral deposition killed all the artificial regeneration; (P. Tollestrup, personal communication, 1995)); all regeneration at the site is natural, which has occurred continuously over the past 15 years, with 4–10-year-old seedlings the most abundant age classes. In the non-scarified sites (all within TB408), fairly continuous regeneration was observed by Tollestrup in all years following logging.

While Tollestrup (1994) concluded that scarification was successful in increasing the numbers of white spruce seedlings, that conclusion is in error. Careful interpretation of the Tollestrup (1994) data indicates that highest densities of regeneration spruce in clearcuts followed natural regeneration onto floodwater silts, moderate densities were found in non-scarified cutblocks, and lowest densities were observed in scarified sites. If the 25-year-old cutblock is removed from comparison, higher seedling densities were observed in the non-scarified sites (median ≈ 1500 spruce ha^{-1}) than in the scarified sites (median ≈ 800 ha^{-1}). The 'scarified' 25-year-old site was scarified twice, after which both artificial regeneration attempts failed. All spruce in the 25-year-old 'scarified' site are natural, postdate scarification by a minimum of 10 years, and follow deposition of floodwater silts (density ≈ 7500 ha^{-1}, stocking $> 80\%$). Lees (1970) found failure of regeneration on scarified bottomlands in Alberta to be common and due to seedbed flooding. Scarification and artificial seeding of spruce on alluvial terrace sites appears to be a risky proposition (failure is common in northern British Columbia alluvial sites also (C. DeLong, personal communication, 1994)).

Density and stocking of white spruce seedlings depends, among other factors, upon seed rain, conditions for germination and establishment, and frequency and magnitude of flooding and siltation (Krasny et al., 1984). An adequate supply of viable seed is the singlemost important determinant of successful natural regeneration (Zasada, 1972). No evidence of increased frost damage or frost mortality of white spruce seedlings was noted in the clearcuts (desiccation was common). Many studies have shown improved white spruce establishment after scarification (e.g. Jarvis et al., 1966; Zasada and Gregory, 1969). Improved stocking, however, must be viewed in context of the ecosystem attributes degraded through site preparation (see Applications, below). Lack of exposed mineral soil in TB408 probably lowered overall stocking and density, but did not cause the regeneration failure. For example, in a 4-year-old cutblock where wardens required the logging company to leave seed islands of canopy dominant spruce (Table 1, Stocking no. 13), seedling + transgressive stocking and density were 68% and 2170 ha^{-1}. In a 27-year-old cutblock, where some residual spruce were left, stocking and density were 89% and 5136 ha^{-1} (Stocking no. 1). In the above two cases, white spruce seed tree distances were 60 and 25 m, respectively.

4.7. Applications to boreal riverine white spruce forests on silty alluvium

For maintenance of biodiversity and ecological function clearcuts are not advisable in a boreal riparian ecosystem characterized by low fire frequency, old-growth structure, and gap dynamics (Perry et al., 1989a; Maser, 1990; Hammond, 1991; Timoney and Robinson, 1996). Eight harvest cutting methods, ranging from clearcut to individual tree selection, were investigated over 10 years in white spruce–aspen forest in Alberta by Lees (1964). He found that selective cutting resulted in highest stand growth rates, lowest mortality, and best regeneration status. Selective or shelterwood methods should be used, leaving abundant healthy seed trees dispersed throughout the cut at a density of ≥ 200 ha^{-1} (corresponding to seedling densities of ≥ 1000 ha^{-1}). Shelterwood cuts should leave 9–14 m^2 ha^{-1} of dominant spruce. Selective cuts should leave 14–23 m^2 ha^{-1} (Zasada, 1972), about one-half the basal area of mature and old-growth spruce in TB408 (K.P. Timoney, unpublished data). Snags should be left standing near pre-logging densities (≈ 50 ha^{-1}) to ensure that: (1) snag dependent mammals and birds are provided for (Cline et al., 1980; Timoney and Robinson, 1996); (2) subsequent recruitment as logs supplies the forest floor with new woody structure (Cline et al., 1980; Morrison and Raphael, 1993; Raphael and Morrison, 1987; Graham et al., 1994).

There are no data to support excessive windthrow of white spruce after partial cutting. In a 10-year study, Lees (1964) found that windthrow in residual stands was slight, and occurred mainly in stems damaged by logging. Assessment of white spruce windthrow over 5 years following partial logging in the Whitecourt, Hinton, and Drayton Valley areas of Alberta indicated ≈ 5–10% windthrow in the first 2 years after partial cutting; no data on windthrow in unlogged replicates were presented for comparison (Navratil et al., 1994). They found that most windthrow occurred in the first 3 years and attributed improved stability to crown, stem, and root growth. After 2 years, Coates (1995) found no evidence that the level of partial cutting (no-harvest, light and heavy removal) affects the extent of windfall in a mixed boreal conifer forest near Date Creek, British Columbia. Windthrow was just as common in unlogged replicates as in those where logging occurred. Factors such as orientation of cutblocks in relation to prevailing winds, and tree slenderness coefficient (height/dbh ratio) may be more important than extent of tree removal (Navratil et al., 1994; Coates, 1995; Steventon, 1995). More study is needed, especially comparison between logging treatments and background levels of windthrow in unmodified forests.

Logging in boreal riparian white spruce should be conducted in winter, both for ease of operations and to protect the forest floor (Corns and Annas, 1986; Timoney and Robinson, 1996). Cutting in poor seed years and in drought years should be avoided as far as possible. If post-harvest survey indicates insufficient stocking and density, we recommend planting seedlings to within 25 m of white spruce seed trees. Rotation length should exceed 200 years, as old-growth attributes begin to appear in riparian white spruce at ≈ 160 years (Timoney and Robinson, 1996). To operate on a rotation shorter than 200 years would remove boreal riparian old-growth forests from the landscape.

From 50% to 75% protection of spruce understory during logging of aspen overstory can be achieved with conventional equipment, and better training, layout, planning, and supervision (Froning, 1980; Brace, 1990; Navratil et al., 1994). Operator training and effort are the most important determinants of understory protection, and these benefits can be achieved with as little as 8% loss of gross productivity (Brace, 1990; Corns, 1991).

Both clearcutting and site preparation should be avoided to protect critical structural and functional attributes of the ecosystem (Franklin, 1990). Riparian forest gaps, snags, nursery logs, advance growth, biodiversity, mycorrhizal symbionts, soil biota, and site productivity are all degraded or lost through clearcutting and site preparation (Cline et al., 1980;

Perry et al., 1989b; Youngblood and Zasada, 1991; Graham et al., 1994; Coates and Burton, 1995; Timoney and Robinson, 1996). Protection of coarse woody debris (logs, snags, etc.) is critical to maintaining forest productivity, and healthy mycorrhizal and microbial communities. In the cutblocks of TB408, there are virtually no snags (median 0 ha^{-1}, 0 m tall, 0 cm dbh, $n = 21$ sites); in undisturbed spruce and spruce–balsam poplar forests, median values are 67 ha^{-1}, 3.5 m tall, 21.0 cm diameter, $n = 17$ sites). Recruitment of logs through falling of snags will be insufficient to supply the future cutblock plant communities with coarse woody debris. Of all site preparation techniques, windrowing and blade scarification cause the most soil degradation in British Columbia (Utzig and Walmsley, 1988). On boreal silty alluvium, plowing and winter ripper plowing are 'especially degrading of site' (C. DeLong, personal communication, 1994). Costs of rehabilitating degraded soils alone range from $500 to 5000 ha^{-1}, and rehabilitation seldom fully restores site productivity; prevention of site degradation is less costly and more effective (Utzig and Walmsley, 1988). Site preparation in riparian environments has the added disadvantage of increasing the risk of competition from undesirable hardwoods and bluejoint reedgrass (*Calamagrostis canadensis* (Michx.) Beauv.) (Corns and Annas, 1986; DeLong et al., 1990; Coates et al., 1994). Fortunately, there has been no scarification in TB408; cover of bluejoint has not risen above natural cover values (0–1%).

The influence of mechanical site preparation on planted white spruce seedlings on silty river floodplains in Alaska has been studied by Youngblood and Zasada (1991). They found little difference in growth and survival of planted containerized spruce regardless of cutting method or site preparation and concluded that adequate growth of spruce may be obtained with partial canopy retention, no site preparation, and planting of locally adapted containerized seedlings. This small difference in growth of white spruce seedlings on scarified (outside TB408) compared to unscarified sites (within TB408) has been corroborated for Peace River alluvium by Tollestrup (1994) who produced regression equations for scarified (height = 1.29(age) + 0.99(age)^2, $R^2 = 0.92$, $n = 402$) and unscarified sites (height = 4.60(age) + 0.51(age)^2, $R^2 = 0.93$, $n = 668$), corresponding to a height at 10 years of 112 cm on scarified sites and 97 cm on unscarified sites.

5. Conclusions

Boreal riparian ecosystems differ fundamentally from boreal uplands. Long fire return interval permits the development of old-growth structure (such as large nursery logs and snags) and function (such as gap phase dynamics, use of tree cavities by animals) that play only minor roles in boreal uplands. The practices of clearcutting and site preparation, which may be appropriate to young, even-aged forests with little biological legacy, are destructive of the attributes that provide health, resilience, and biodiversity to boreal riparian forests. Managers, ecologists, and the public must soon realize that clearcutting and site preparation are inappropriate to boreal riparian ecosystems. If we fail to do so, we will be left with tree plantations with little potential for restoration of old-growth attributes.

In boreal riparian ecosystems dominated by white spruce, forestry companies may avoid the costs of site preparation, minimize costs of artificial regeneration, and maintain long-term forest productivity and biodiversity by adopting many of the above recommendations. The future of boreal riparian forestry, if there is to be a future, lies in a shift from forest removal and plantation establishment to removal of trees while leaving behind a functional forest ecosystem, one that provides both timber and non-timber values.

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